MAPÚA MALAYAN DIGITAL COLLEGE
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BIOMECHANICS BREAKDOWN

Biomechanics Of Pickleball

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Overview: Biomechanics of Pickleball

COURT CONTEXT & DEMANDS

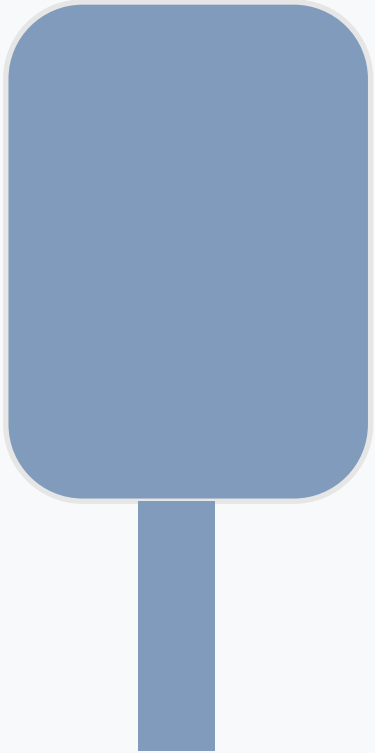
- Pickleball combines multi-directional agility, low center-of-mass control at the Kitchen line, and explosive overhead striking.

BIOMECHANICAL CORE

- Power is generated through a bottom-up kinetic chain — starting from ground reaction forces (GRF), transferring through hip/core rotation, and terminating at the paddle impact point.

ANALYSIS FOCUS

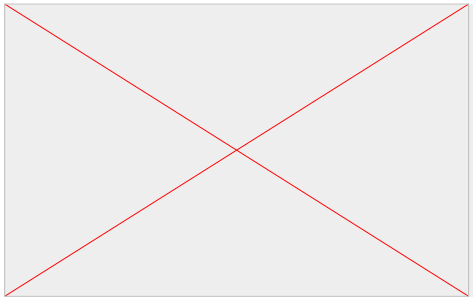
- Evaluating movement mechanics, muscle activation, and kinetic loads across three foundational skills to maximize performance and minimize injury.



Movements and Actions

- Forehand Drive
- The Overhead Smash
- Dink Shot & Split Step

MOVEMENT/ACTION



Forehand Drive

- Preparation/Backswing: unit turn, shoulder rotation, weight transfer to the back foot.
- Acceleration: hips and torso uncoil, paddle swings low-to-high.
- Impact: contact made in front of the lead hip with a stable wrist.
- Follow-through: paddle sweeps across the body, weight fully transfers forward.



The Overhead Smash

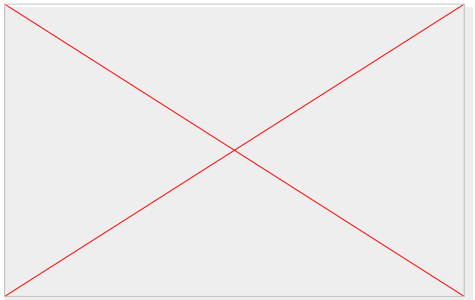
- Preparation: rapid backward tracking, split-step setup, arm cocks into a 'trophy position.'
- Drive phase: explosive leg push-off, hip-shoulder separation, upward chest extension.
- Impact: full overhead extension with a forceful downward wrist snap.
- Landing: eccentric deceleration onto the lead foot.



Dink Shot & Split Step

- Split step: low-amplitude hop as the opponent strikes, landing with a wide base of support.
- Dink execution: low-to-high soft stroke from the shoulder, minimal wrist motion.
- Recovery: instant return to an athletic stance at the Kitchen line.

BIOMECHANICS



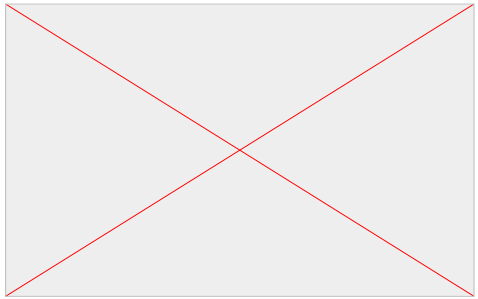
Forehand Drive

- Lever class: 3rd class lever — fulcrum at the shoulder, effort from Pectoralis/Deltoid, load at the paddle/ball impact.
- Kinetic chain: GRF → knee extension → pelvic rotation → torso rotation → shoulder/arm extension.



The Overhead Smash

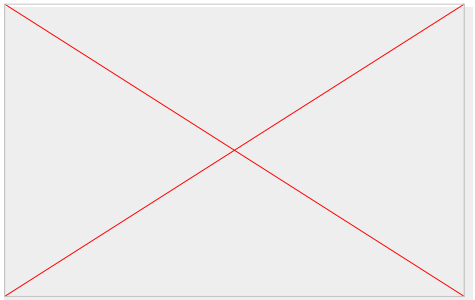
- Stretch-Shortening Cycle (SSC): rapid pre-stretch of the chest and shoulder internal rotators yields an explosive elastic contraction.
- Rotational acceleration: extreme internal shoulder rotation velocity maximizes downward ball trajectory.



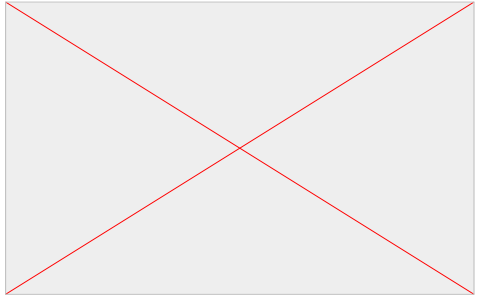
Dink Shot & Split Step

- Center of gravity (COG): kept low and inside the base of support (BOS) for reactive stability.
- Impulse-momentum relationship: a soft paddle touch absorbs incoming ball energy rather than adding velocity.

KINEMATIC or KINEMATIC MEASURES







Forehand Drive

- Wrist extension at backswing: ~20°-30°.
- Peak angular paddle speed: ~15-20 m/s.
- Peak GRF: ~1.5x-2.0x body weight during plant-and-drive.
- Torso rotational torque transferred through core stabilizers.

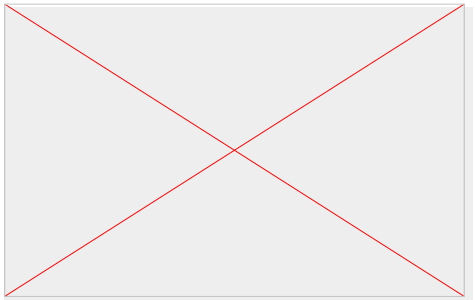
The Overhead Smash

- Shoulder elevation angle at contact: 160°-180°.
- Peak internal shoulder rotational velocity: >1000° /sec.
- Vertical ground impulse during push-off.
- Significant shoulder joint shear force during high-speed deceleration.

Dink Shot & Split Step

- Joint angle: deep knee flexion (60°-90° hip/knee bend) during low dink retrieval.
- Minimal paddle acceleration (<5 m/s).

MUSCLE ACTIVATION



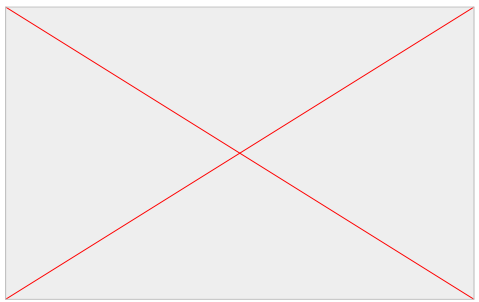
Forehand Drive

- Agonists: Pectoralis Major, Anterior Deltoid, Rectus Femoris (leg drive).
- Synergists/Stabilizers: rotator cuff (Supraspinatus, Infraspinatus), Rectus Abdominis, obliques, forearm flexors.



The Overhead Smash

- Agonists: Latissimus Dorsi, Pectoralis Major, Triceps Brachii, Gastrocnemius/Soleus.
- Stabilizers: rotator cuff (prevents shoulder dislocation on deceleration), core abdominis (limits lumbar hyperextension).



Dink Shot & Split Step

- Quadriceps (isometric/eccentric load), Gluteus Medius (lateral stabilization).
- Soleus, forearm extensors.

Injury Risk Analysis & Mitigation

Injury Risk	Biomechanical Cause	Prevention & Mitigation
Lateral Epicondylitis ("Pickleball Elbow")	Repetitive wrist-extension overload and off-center paddle impacts.	Maintain a stiff wrist during dinks; hit in front of the body using the full arm/shoulder unit.
Achilles Tendon Strain / Tear	Sudden backward tracking or forceful plant-and-cut acceleration without warm-up.	Turn-and-run footwork instead of backpedaling; dynamic calf warm-ups.
Rotator Cuff Tendinopathy	High-velocity repetitive overhead smashes without adequate leg drive.	Engage the full kinetic chain (legs + core rotation) to relieve shoulder joint overload.
Ankle Sprains	Rapid lateral shuffling and catching a shoe edge on non-marking court surfaces.	Practice proper split-step landings; wear court-specific lateral-support footwear.

Physiological Considerations

ENERGY SYSTEMS

ATP-PCr (0-10 sec) immediate energy for explosive rallies, split-steps, smashes

Glycolytic (10-60 sec) sustained, intense Kitchen dink exchanges

Aerobic (1-2 hrs) baseline recovery between rallies, overall match endurance

NEUROMUSCULAR & CARDIORESPIRATORY DEMAND

- Eccentric quadriceps fatigue builds during sustained low-crouch Kitchen play.
- Reactive agility and fast-twitch motor unit recruitment are required for close-range dink battles (<0.4 sec reaction window).

ENERGY DEMAND SPLIT



Conclusion & Practical Summary

KEY TAKEAWAYS

- Kinetic chain first: power originates from ground reaction forces — relying solely on the arm increases injury risk at the elbow and shoulder.
- Low center of gravity: knee flexion and a wide base of support at the Kitchen line optimize balance, deceleration, and reaction time.
- Injury prevention strategy: combine kinetic-chain technique, court-specific footwear, and targeted rotator-cuff/core conditioning for long-term play longevity.